

ENGR 225: Mechanics 1

Homework Coversheet

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ASSIGNMENT: Homework 2

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

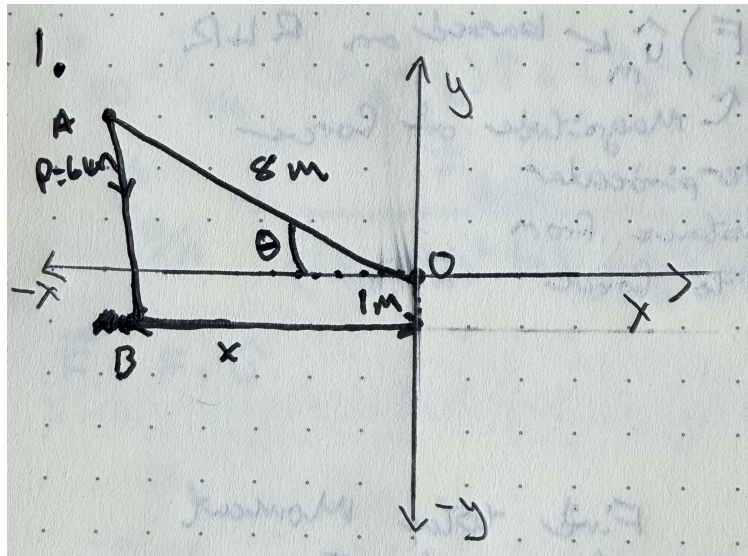
- ☒ None
- ☐ Minor
- ☐ Major
- ☐ Extensive

Mark the box that applies (e.g., replace ☐ with ☒, or type an X beside it, before compiling/printing). Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

Problem 1

Given / Find: A crane boom OA (length 8 m) makes an angle $\theta = 30^\circ$ with the horizontal. A towline force $P = 6$ kN acts at A , directed toward a hook at B on the ground, a horizontal distance x from the point below O (O is 1 m above the ground). Find the placement x that maximizes the moment of P about O , and find that maximum moment.

Sketch:



Equations & Solution:

Since A and O are fixed, the moment of a fixed-magnitude P applied at A is largest when the line of action AB is *perpendicular* to OA :

$$M_{\max} = P(OA)$$

Place O at the origin, x pointing left (toward B) and y up:

$$\vec{r}_{OA} = (8 \cos 30^\circ, 8 \sin 30^\circ) = (6.928, 4.000) \text{ m}, \quad \vec{r}_{AB} = (x - 6.928, -1 - 4.000)$$

Perpendicularity requires $\vec{r}_{OA} \cdot \vec{r}_{AB} = 0$:

$$6.928(x - 6.928) + 4.000(-5.000) = 0$$

Solve:

$$6.928x = 6.928^2 + 20.0 = 48.0 + 20.0 = 68.0 \Rightarrow x = \frac{68.0}{6.928} = 9.81 \text{ m}$$

$$M_{\max} = P(OA) = (6 \text{ kN})(8 \text{ m}) = 48.0 \text{ kN} \cdot \text{m}$$

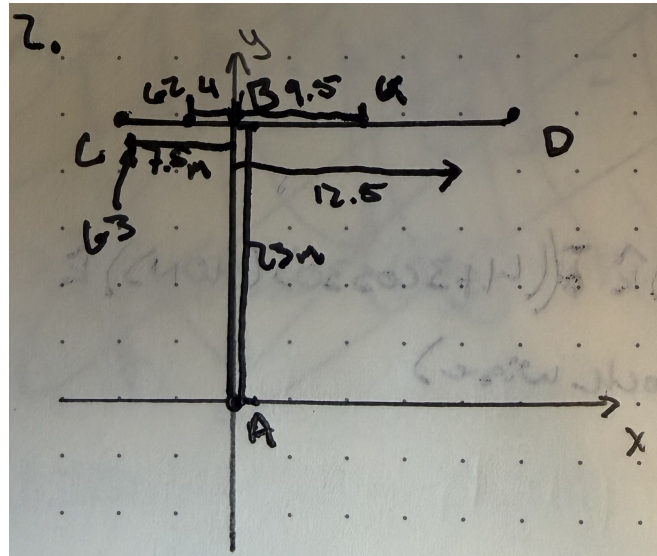
Answer

$$x = 9.81 \text{ m}, \quad M = 48.0 \text{ kN} \cdot \text{m}$$

Problem 2

Given / Find: A tower crane has a jib BD (1.5 Mg, mass center G_1 , 9.5 m right of the tower) carrying a 2 Mg load (12.5 m right of tower), and a counter-jib BC (0.5 Mg, mass center G_2 , 4 m left of tower) with counterweight C (mass center G_3 , 7.5 m left of tower). Find the counterweight mass m so the net moment of all these weights about the base A is zero.

Sketch:



Equations & Solution:

$$\sum M_A = 0 : \quad (0.5 \text{ Mg})(4 \text{ m}) + m(7.5 \text{ m}) = (1.5 \text{ Mg})(9.5 \text{ m}) + (2 \text{ Mg})(12.5 \text{ m})$$

Solve:

$$\begin{aligned} 2.0 + 7.5m &= 14.25 + 25.0 = 39.25 \\ m &= \frac{39.25 - 2.0}{7.5} = \frac{37.25}{7.5} = 4.97 \text{ Mg} \end{aligned}$$

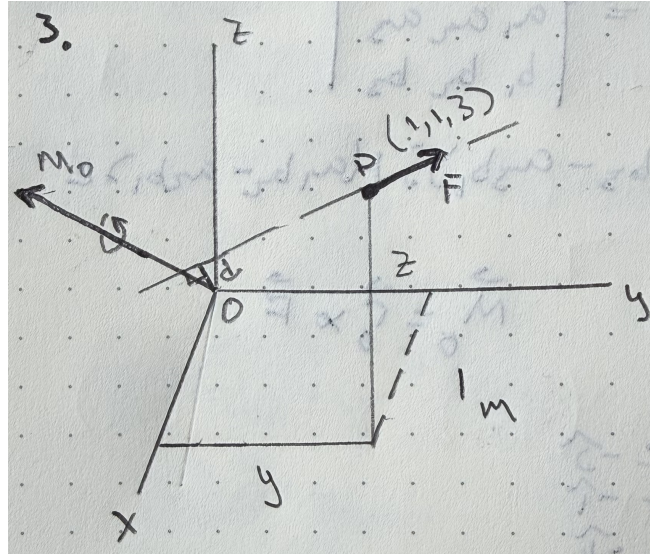
Answer

$$m = 4.97 \text{ Mg}$$

Problem 3

Given / Find: $\vec{F} = 6\hat{i} + 8\hat{j} + 10\hat{k}$ N produces a moment $\vec{M}_O = -14\hat{i} + 8\hat{j} + 2\hat{k}$ N·m about O . The line of action passes through a point with $x = 1$ m. Find y, z of that point, and the perpendicular distance d from O to the line of action.

Sketch:



Equations & Solution:

The point lies at $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ with $\vec{M}_O = \vec{r} \times \vec{F}$:

$$\vec{r} \times \vec{F} = (10y - 8z)\hat{i} - (10x - 6z)\hat{j} + (8x - 6y)\hat{k}$$

Matching components to $\vec{M}_O = (-14, 8, 2)$ with $x = 1$:

$$10y - 8z = -14, \quad -(10(1) - 6z) = 8, \quad 8(1) - 6y = 2$$

Solve: From the $\hat{j}$ -equation: $-10 + 6z = 8 \Rightarrow z = 3.00$ m. From the $\hat{k}$ -equation: $8 - 6y = 2 \Rightarrow y = 1.00$ m (check $\hat{i}$: $10(1) - 8(3) = -14$ ✓).

For the perpendicular distance, since $|\vec{M}_O| = |\vec{F}| d$:

$$d = \frac{|\vec{M}_O|}{|\vec{F}|} = \frac{\sqrt{14^2 + 8^2 + 2^2}}{\sqrt{6^2 + 8^2 + 10^2}} = \frac{\sqrt{264}}{\sqrt{200}} = \frac{2\sqrt{66}}{10\sqrt{2}} = \frac{\sqrt{33}}{5} \approx 1.15 \text{ m}$$

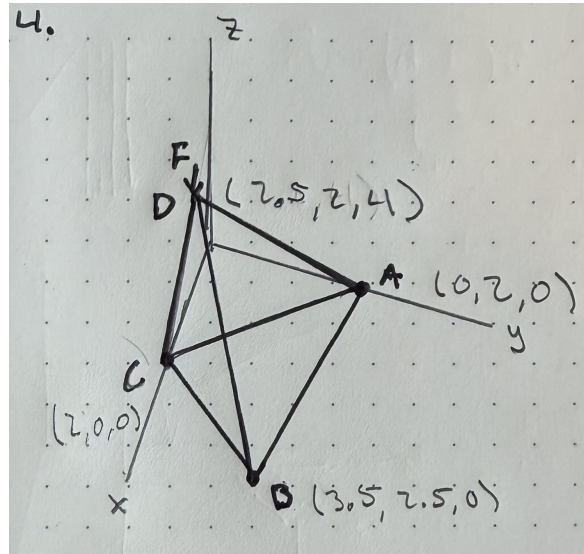
Answer

$$\vec{r} = 1\hat{i} + 1\hat{j} + 3\hat{k} \text{ m} \quad (y = 1.00 \text{ m}, z = 3.00 \text{ m}), \quad d = 1.15 \text{ m}$$

Problem 4

Given / Find: A tripod has base points B, C, A and apex D , with $\vec{F} = 50\hat{i} - 20\hat{j} - 80\hat{k}$ N applied at D . Using the given dimensions, $A = (0, 2, 0)$ m, $B = (3.5, 2.5, 0)$ m, $C = (2, 0, 0)$ m, and $D = (2.5, 2, 4)$ m. Find the magnitude of the moment of $\vec{F}$ about line CA .

Sketch:



Equations & Solution:

Compute the moment of $\vec{F}$ about point A , then project onto the unit vector along CA :

$$\vec{r}_{AD} = \vec{r}_D - \vec{r}_A, \quad \vec{M}_A = \vec{r}_{AD} \times \vec{F}, \quad \hat{u}_{CA} = \frac{\vec{r}_A - \vec{r}_C}{|\vec{r}_A - \vec{r}_C|}, \quad M_{CA} = \hat{u}_{CA} \cdot \vec{M}_A$$

Solve:

$$\vec{r}_{AD} = (2.5, 0, 4.0) \text{ m}$$

$$\vec{M}_A = \vec{r}_{AD} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2.5 & 0 & 4.0 \\ 50 & -20 & -80 \end{vmatrix} = (80)\hat{i} + (400)\hat{j} + (-50)\hat{k} \text{ N} \cdot \text{m}$$

$$\hat{u}_{CA} = \frac{(-2, 2, 0)}{2.828} = (-0.7071, 0.7071, 0)$$

$$M_{CA} = \hat{u}_{CA} \cdot \vec{M}_A = (-0.7071)(80) + (0.7071)(400) + 0 = 226.3 \text{ N} \cdot \text{m}$$

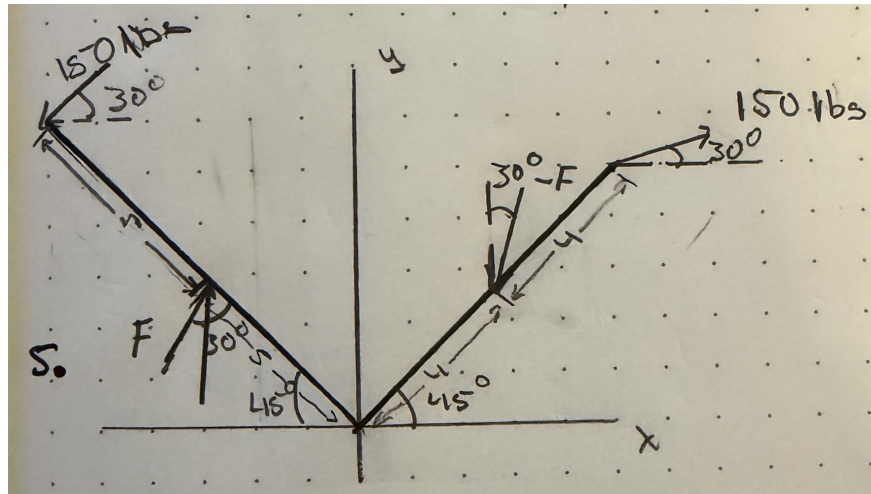
Answer

$$M_{CA} = 226 \text{ N} \cdot \text{m}$$

Problem 5

Given / Find: A symmetric crank pivoted at O has two straight arms at 45° to the horizontal. On each arm a rod is pinned at 4 in from O (carrying $F / -F$, each 30° from vertical) and at 9 in from O (carrying the 150 lb forces, each 30° from horizontal). Find F so the net couple moment on the crank is zero.

Sketch:



Equations & Solution:

Each pair of forces is equal, opposite, and parallel, so each pair is a pure couple. For a couple, $M = Fd$ with d the perpendicular distance between the two lines of action, independent of the point taken.

The two pins of a pair sit symmetrically on the 45° arms, so their separation is horizontal:

$$s_{150} = 2(9 \cos 45^\circ) = 9\sqrt{2} \text{ in}, \quad s_F = 2(4 \cos 45^\circ) = 4\sqrt{2} \text{ in}$$

The 150 lb forces are 30° from horizontal and the F forces are 30° from vertical, so

$$d_{150} = 9\sqrt{2} \sin 30^\circ, \quad d_F = 4\sqrt{2} \cos 30^\circ$$

Solve: The two couples must cancel, so $150 d_{150} = F d_F$:

$$F = \frac{150 (9\sqrt{2} \sin 30^\circ)}{4\sqrt{2} \cos 30^\circ} = \frac{1350}{4} \tan 30^\circ = 194.9 \text{ lb}$$

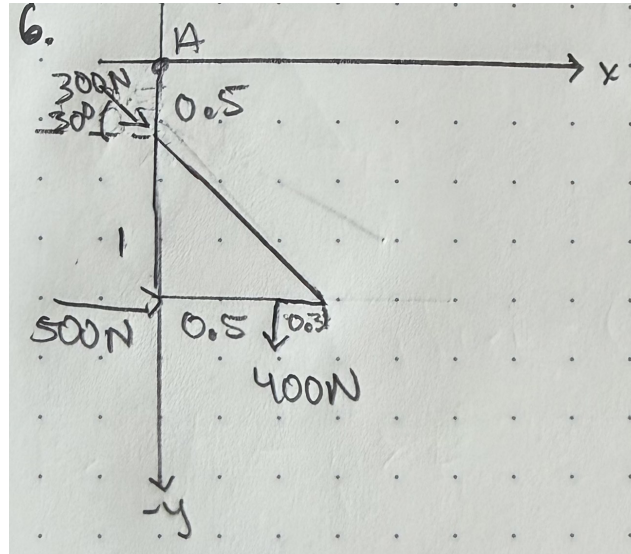
Answer

$$F = 195 \text{ lb}$$

Problem 6

Given / Find: A frame is pinned at A . Forces act on it: 300 N at 30° from horizontal (0.5 m below A), 500 N horizontal (1.5 m below A), and 400 N vertical (0.5 m to the right, along the base). Replace this system with an equivalent resultant force and couple moment at A .

Sketch:



Equations & Solution:

Let x be right, y up, origin at A .

$$\vec{F}_1 = 300(\cos 30^\circ, -\sin 30^\circ), \quad \vec{F}_2 = (500, 0), \quad \vec{F}_3 = (0, -400)$$

$$\vec{F}_R = \sum \vec{F}_i, \quad M_A = \sum (x_i F_{yi} - y_i F_{xi})$$

with application points $\vec{r}_1 = (0, -0.5)$, $\vec{r}_2 = (0, -1.5)$, $\vec{r}_3 = (0.5, -1.5)$ m.

Solve:

$$\vec{F}_R = (300 \cos 30^\circ + 500, -300 \sin 30^\circ - 400) = (759.8, -550.0) \text{ N}$$

$$|F_R| = \sqrt{759.8^2 + 550.0^2} = 938 \text{ N}, \quad \theta = \tan^{-1}\left(\frac{550.0}{759.8}\right) = 35.9^\circ \text{ below } x\text{-axis}$$

$$M_A = [0(-150.0) - (-0.5)(259.8)] + [0(0) - (-1.5)(500)] + [0.5(-400) - (-1.5)(0)]$$

$$M_A = 129.9 + 750.0 - 200.0 = 679.9 \text{ N} \cdot \text{m, CCW}$$

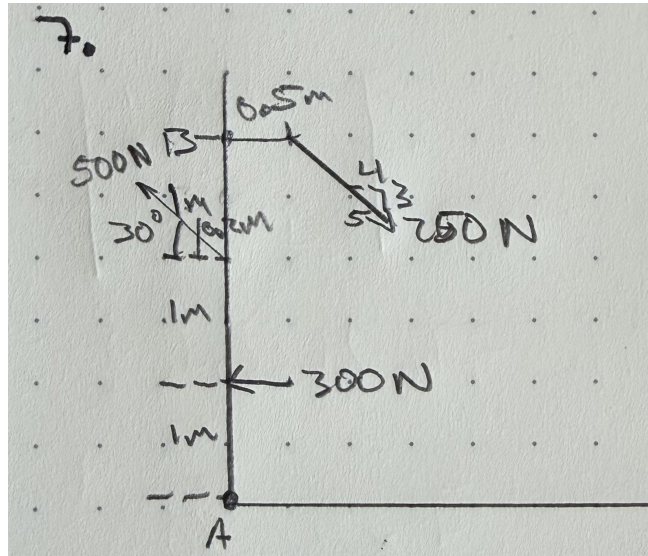
Answer

$$F_R = 938 \text{ N}, \quad \theta = 35.9^\circ \text{ below the } x\text{-axis}, \quad M_A = 680 \text{ N} \cdot \text{m (CCW)}$$

Problem 7

Given / Find: A post AB (3 m tall) carries a 500 N cable (1 m below B , 30° from horizontal, offset 0.2 m from centerline), a 250 N cable at B (3-4-5 triangle, offset 0.5 m), and a 300 N horizontal force (2 m below B). Replace with a single resultant force and find where its line of action crosses post AB , measured from B .

Sketch:



Equations & Solution:

With B at the origin, x right, y up:

$$\vec{F}_1 = 500(-\cos 30^\circ, \sin 30^\circ) \text{ at } (-0.2, -1), \quad \vec{F}_2 = 250\left(\frac{4}{5}, -\frac{3}{5}\right) \text{ at } (0.5, 0), \quad \vec{F}_3 = (-300, 0) \text{ at } (0, -2)$$

$$\vec{F}_R = \sum \vec{F}_i, \quad M_B = \sum (x_i F_{yi} - y_i F_{xi}), \quad r_b = \frac{M_B}{F_{Rx}}$$

Solve:

$$\vec{F}_R = (-500 \cos 30^\circ + 200 - 300, 500 \sin 30^\circ - 150) = (-533.0, 100.0) \text{ N}$$

$$M_B = [(-0.2)(250) - (-1)(-433.0)] + [(0.5)(-150) - 0] + [0 - (-2)(-300)]$$

$$M_B = -1158.0 \text{ N} \cdot \text{m}, \quad r_b = \frac{-1158.0}{-533.0} = 2.17 \text{ m below } B$$

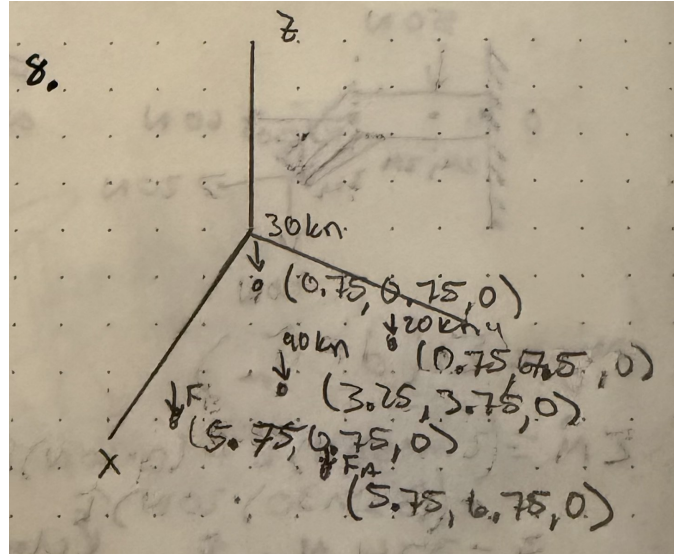
Answer

$$\vec{F}_R = -533\hat{i} + 100\hat{j} \text{ N}, \quad \text{line of action crosses } AB \text{ at } r_b = 2.17 \text{ m below } B$$

Problem 8

Given / Find: A rectangular concrete slab carries five downward point loads: $F_B = 35$ kN, $F_A = 40$ kN, a 30 kN post, a 90 kN post, and a 20 kN post, at the (x, y) locations shown. Find the resultant force and the (x, y) location of its point of application.

Sketch:



Equations & Solution:

All five forces are parallel (downward, $-\hat{k}$), so the resultant is their scalar sum, and its line of action passes through the weighted centroid of the application points:

$$\vec{F}_R = -\sum F_i \hat{k}, \quad \bar{x} = \frac{\sum F_i x_i}{\sum F_i}, \quad \bar{y} = \frac{\sum F_i y_i}{\sum F_i}$$

The origin is the far corner, x along the 6.5 m edge and y along the 7.5 m edge. Every post is inset 0.75 m from its nearest edges and the center lines fall at $x = 3.25$, $y = 3.75$, so the loads act at 30: (0.75, 0.75), F_B : (5.75, 0.75), 90: (3.25, 3.75), F_A : (5.75, 6.75), 20: (0.75, 6.75) m.

Solve:

$$\sum F_i = 30 + 35 + 90 + 40 + 20 = 215 \text{ kN}$$

$$\bar{x} = \frac{30(0.75) + 35(5.75) + 90(3.25) + 40(5.75) + 20(0.75)}{215} = \frac{761.25}{215} = 3.54 \text{ m}$$

$$\bar{y} = \frac{30(0.75) + 35(0.75) + 90(3.75) + 40(6.75) + 20(6.75)}{215} = \frac{791.25}{215} = 3.68 \text{ m}$$

Answer
$\vec{F}_R = -215\hat{k} \text{ kN}, \quad (x, y) = (3.54, 3.68) \text{ m}$